

Assignment 1: E-Commerce Sales Analytics

Scenario: An e-commerce company stores transactional data in MySQL and wants to move historical sales data to Hadoop for analytics.

MySQL tables: customers(customer_id, name, city, state, join_date), products(product_id, product_name, category, price), and orders(order_id, customer_id, product_id, quantity, order_date, order_status).

Perform following tasks:

1. Create the database ecommerce_db and all three tables in MySQL. Insert at least **30 customers, 15 products, and 100 orders**.
2. Use **Sqoop** to import all three tables into separate directories under /me/ecommerce/ in HDFS.
3. Verify imported data using appropriate HDFS commands and report the number of files and records.
4. Create a Hive database ecommerce_dw and suitable tables over the imported data.
5. Write HiveQL queries to determine:
 - Total sales revenue.
 - Revenue by product category.
 - Top 5 customers by spending.
 - Top 5 products by quantity sold.
 - Monthly sales revenue.
 - City with the highest sales.
 - Percentage of cancelled orders.
6. Store the result of the individual query into separate file in HDFS .
7. Transfer back these result files to local system.

Assignment 2: University Student Performance Analytics

Scenario: A university maintains student, course, enrollment, and examination information in MySQL. The university wants to analyze academic performance using Hadoop.

MySQL tables: students(student_id, student_name, department, admission_year, gender), courses(course_id, course_name, department, credits), and results(result_id, student_id, course_id, semester, marks, attendance).

Perform following tasks:

1. Create university_db in MySQL and populate it with at least **40 students, 10 courses, and 150 result records**.
2. Import the tables using Sqoop into /me/university/.
3. Import results using **multiple Sqoop mappers** and explain the purpose of --split-by.
4. Inspect the resulting HDFS directories and files.
5. Create appropriate Hive tables.
6. Write Hive queries to find:
 - Average marks for each department.
 - Top 3 students in each department.
 - Pass percentage for each course.
 - Students having attendance below 75%.
 - Relationship between attendance and average marks.
 - Course having the highest failure rate.
 - Semester-wise average performance.
7. Store the result of the individual query into separate file in HDFS .
8. Transfer back these result files to local system.

Assignment 3: Banking Transaction and Customer Analytics

Scenario: A bank stores customer accounts and transactions in MySQL. Its analytics team needs a Hadoop-based solution for transaction analysis.

MySQL tables: customers(customer_id, customer_name, city, account_type, account_open_date), accounts(account_id, customer_id, balance, branch), and transactions(transaction_id, account_id, transaction_type, amount, transaction_date).

Populate the database with at least **30 customers, 40 accounts, and 200 transactions**.

Tasks:

1. Create and populate bank_db.
2. Perform a **full Sqoop import** of customers, accounts and transactions.
3. Verify through HDFS that the new data has been imported.
4. Create Hive tables and answer:
 - Total deposits and withdrawals.
 - Customers with the highest transaction values.
 - Branch-wise account balances.
 - Average transaction amount by account type.
 - Month-wise transaction volume.
 - Accounts with no transactions.
 - Customers whose withdrawal amount exceeds their deposit amount.
5. Store the result of the individual query into separate file in HDFS .
6. Transfer back these result files to local system.

Assignment 4: Hospital Operations Analytics

Scenario: A multi-specialty hospital wants to analyze patient visits, doctors, departments, and billing information stored in MySQL.

MySQL tables: patients(patient_id, patient_name, city, age, gender), doctors(doctor_id, doctor_name, specialization, department), visits(visit_id, patient_id, doctor_id, visit_date, diagnosis, consultation_fee), and payments(payment_id, visit_id, payment_mode, amount, payment_status).

Students should create at least **40 patients, 12 doctors, 150 visits, and 150 payment records** using synthetic data only.

Tasks:

1. Create and populate hospital_db.
2. Import the four tables using Sqoop.
3. Store them under a structured HDFS hierarchy such as /me/hospital/patients, /doctors, /visits, and /payments.
4. Create Hive tables.
5. Use Hive joins and aggregations to determine:
 - Number of visits per department.
 - Revenue generated by each department.
 - Top 5 doctors based on number of consultations.
 - Average consultation fee by specialization.
 - Most common diagnoses.
 - City-wise patient counts.
 - Percentage of successful versus pending payments.
 - Month having the highest number of visits.
6. Store the result of the individual query into separate file in HDFS .
7. Transfer back these result files to local system.

Assignment 5: Retail Data Warehouse — Complete ETL Challenge

Scenario: A retail chain has multiple stores. Operational information is maintained in MySQL, while management wants a Hadoop/Hive analytical warehouse. This assignment combines the concepts from the previous four and is the most advanced.

MySQL tables: stores(store_id, store_name, city, state), customers(customer_id, customer_name, city, customer_segment), products(product_id, product_name, category, unit_price), and sales(sale_id, store_id, customer_id, product_id, quantity, discount, sale_date).

Create at least **5 stores, 50 customers, 20 products, and 250 sales transactions**.

Tasks:

1. Design and create retail_data in MySQL with appropriate primary and foreign keys.
2. Populate realistic synthetic data, including some NULL values and discounted transactions.
3. Import dimension tables using Sqoop.
4. Store the imported datasets in an organized HDFS directory structure.
5. Create a Hive database retail_dw and tables.
6. Perform necessary data cleaning/transformation in Hive, including handling NULLs and calculating:

$\text{gross_amount} = \text{quantity} \times \text{unit_price}$

$\text{net_amount} = \text{gross_amount} \times (1 - \text{discount}/100)$

8. Write analytical queries for:
 - Store-wise revenue.
 - State-wise revenue.
 - Category-wise revenue.
 - Top 10 customers.
 - Top 5 products.
 - Monthly sales trends.

- Average order value by customer segment.
- Store contributing the highest percentage of total revenue.
- Products never sold.
- Month-over-month revenue growth.